

Farmers Insurance – Case Study

Problem Statement:

The PMFBY scheme provides financial security to farmers. We will analyze its effectiveness using structured data analysis. We will use SQL queries to extract key insights on premiums, claims, and farmer demographics. The challenge is to process multi-year data efficiently, identify trends, and generate reports that help stakeholders assess the scheme's impact

Objective:

The focus is on deriving insights from the PMFBY dataset to assess insurance coverage trends, claim settlements, and demographic distributions.

Insurance coverage trends :-

Year	Insurance units	Total Farmers Covered	Insured Land Area	Sum Insured	Gross Premium Amt To Be Paid
2018	141,104	15,022,576	19,886	8,816,201	726,273
2019	138,733	12,354,767	19,324	8,799,657	921,374
2020	122,507	12,169,010	20,130	8,291,459	1,341,142
2021	85,477	10,750,659	15,842	6,906,766	1,311,883
% change 2021 vs 2018	-39%	-28%	-20%	-22%	81%

2021 vs 2018. Year on Year Insurance units had decreased by 39%, Total Farmers Covered, Insured land and sum insured are also decreased. Majorly Premium Amount has increased by 81%.

State-Level Analysis:

For year 2018, State with most extensive insurance coverage are as below: -

State	Total Farmers Covered	Total Sum Insured	Farmers' Premium Amount
MAHARASHTRA	2,126,801	4,495,200	85,274
UTTAR PRADESH	2,028,806	1,003,506	19,252
MADHYA PRADESH	1,529,864	1,421,496	22,002
RAJASTHAN	1,508,808	1,332,691	24,577
HARYANA	613,509	697,107	10,435

Key State-Level Insights

- **Top States by Coverage:** Maharashtra, Uttar Pradesh, and Madhya Pradesh lead in terms of the number of farmers covered under insurance schemes.

- **Highest Sum Insured:** Maharashtra stands out with the highest total sum insured, which is significantly larger than any other state. This suggests not only a high number of farmers but also potentially higher-value crops being insured.
- **Premium vs. Sum Insured:** The premium as a percentage of the sum insured varies considerably. States like West Bengal (6.88%) and Assam (7.13%) have a relatively high premium rate compared to states like Madhya Pradesh (1.55%) and Haryana (1.50%). This could be due to the type of crops insured, perceived risk levels, or different subsidy structures.

District-Level Analysis

District	State	Total Farmers Covered	Total Sum Insured	Farmers' Premium
Bid	MAHARASHTRA	387,806	940,876	1,979
Osmanabad	MAHARASHTRA	258,677	722,732	1,902
Ahmadnagar	MAHARASHTRA	230,923	64,970	1,433
Purba Medinipur	WEST BENGAL	283,591	64,577	3
Paschim Medinipur	WEST BENGAL	283,245	104,678	1,654

Key District-Level Insights

1. **Maharashtra's Dominance:** Districts from Maharashtra, particularly **Bid** and **Osmanabad**, lead significantly in both the number of farmers covered and the total sum insured. This highlights the extensive reach and high value of agricultural insurance in this region.
2. **High Participation in West Bengal:** Districts in West Bengal, such as **Purba Medinipur** and **Paschim Medinipur**, show a very high number of farmers covered. However, the average sum insured per farmer is much lower compared to districts in Maharashtra. This could indicate that while the insurance schemes are widespread, they may cover smaller land holdings or lower-value crops
3. **Madhya Pradesh's Strong Showing:** Several districts in Madhya Pradesh, including **Ujjain**, **Sehore**, and **Dewas**, have a large number of insured farmers. The average sum insured per farmer in these districts is robust.

Coverage vs. Population

State	Total Farmers Covered	Total Population	Rural Population	% of Total Population Covered	% of Rural Population Covered
MAHARASHTRA	2,126,801	112,374,333	61,556,074	1.89%	3.45%
UTTAR PRADESH	2,028,806	199,812,341	155,317,278	1.02%	1.31%
MADHYA PRADESH	1,529,864	72,626,809	52,557,404	2.11%	2.91%
RAJASTHAN	1,508,808	68,548,437	51,500,352	2.20%	2.93%
HARYANA	613,509	25,351,462	16,509,359	2.42%	3.72%

1. Reach of Insurance Relative to Population:

- While states like Maharashtra and Uttar Pradesh have a high absolute number of farmers covered, when viewed as a percentage of their large rural populations, the penetration rate is still relatively low (3.45% and 1.31% respectively). This suggests that even in states with high coverage, a large portion of the rural population may not be insured under these schemes.
- Haryana stands out with a higher percentage of its rural population covered (3.72%) compared to other states in the top 10, indicating a more effective reach of insurance programs relative to its population size.
- States like Odisha and Telangana, despite having significant rural populations, show lower coverage rates, suggesting that there is considerable scope for expanding insurance penetration in these areas.

% Percentage Distribution of Farmers Covered

SC	ST	OBC	General
10.32 %	6.49 %	38.64 %	44.55 %

Male	Female
85.23 %	14.69 %

Conclusion:

PFMBY need to focus to increase the insurance units and farmers covered under these schemes, need to conduct awareness session for farmers to understand the benefit of PMFBY. Use digital platform / biometric to conduct paperless registration for farmers. Involving Bank for transactions and SMS notification and for each notification in Unicode language SMS messages. Faster claim settlement process for the farmers. Use technology for weather forecast and agriculture education to the farmers.